

N6Cam — Test Report

Date: 2026-05-23 16:58:30
Device: SIANA N6Cam (STM32N657 / IMX335 / Neural-ART NPU)
Firmware: 01.08.2593089169
MCU UID: 003031564236500400420024
Shell TTY: /dev/ttyACM1
Runtime: 65 seconds

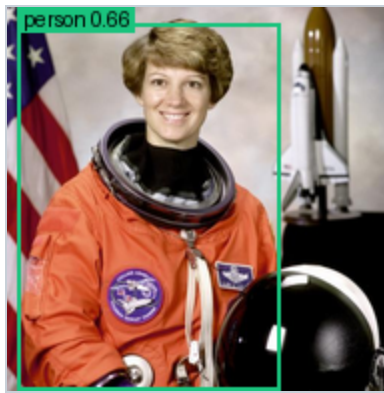
Result: ALL TESTS PASSED — Total: 56 | Pass: 55 | Fail: 0 | Skip: 1

Test ID	Description	Result
T00.1	Kit CDC port /dev/ttyACM1 exists and is readable	PASS
T00.2	Shell `help` lists user-defined commands	PASS
T00.3	`fw` returns FW VER: ...	PASS
T00.4	`uid` returns MCU UID	PASS
T00.5	`uptime` returns UPTIME ms	PASS
T01.1	`echo query` reports state + ok ack	PASS
T01.2	`echo off` succeeds (ok-ack)	PASS
T01.3	After `echo off`, no echoed prompt before reply	PASS
T01.4	`echo on` restores echo + ok ack	PASS
T02.1	`rtc` returns DD/MM/YY HH:MM:SS	PASS
T02.2	`rtc set DDMMYYYYHHMMSS` accepts a valid value	PASS
T02.3	RTC reads back the value we just set	PASS
T02.4	`version` prints Application:	PASS
T03.1	`irled on` reports state=1 + ok-ack	PASS
T03.2	`irled off` reports state=0	PASS
T03.3	`motion sense 65 45` accepted, echoed back	PASS
T03.4	`motion query` returns persisted 65/45	PASS

T04.1	`img quality 75` set + ok-ack	PASS
T04.2	`img color RGB` accepted	PASS
T04.3	`img chroma 1` accepted	PASS
T04.4	`img query` shows quality=75 color=RGB chroma=1	PASS
T05.1	`detect start` succeeds	PASS
T05.2	`detect profile 3 1` (people+vehicles, save-SD)	PASS
T05.3	`detect profile query` returns 0x03 / 0x01	PASS
T06.1	`notify enable 0x10` (people-detect bit) accepted	PASS
T06.2	`notify period 30` accepted	PASS
T06.3	`notify query` shows mask=0x10 period=30	PASS
T06.4	`notify trigger 16` emits +SDVRNTF JSON with rsn=16	PASS
T06.5	`notify disable` succeeds	PASS
T07.1	`sd query` reports mounted	PASS
T07.2	`sd ls` returns at least one entry	PASS
T07.3	`photo savesd` returns SoW §7 filename 4325412_23052026_165732.rdy	PASS
T07.4	New .rdy file appears on SD after `photo savesd` 4325412_23052026_165732.rdy	PASS
T08.1	`frame clear` reverts to live camera	PASS
T08.2	Upload a synthetic 192x192 grey frame	PASS
T08.3	`frame query` reports loaded	PASS
T08.4	`frame run` returns a count + NN time on a grey frame count=0 NN=25.1ms	PASS
T09.1	astronaut.jpg (real person) → NN detects ≥1 person count=1 NN=25.1ms	PASS



input (192×192)



kit detections

camera.jpg (real person) → NN detects ≥ 1 person
count=1 NN=25.1ms

T09.2



input (192×192)

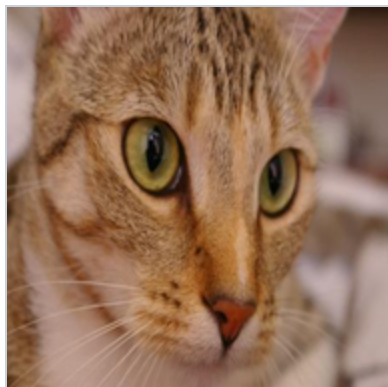


kit detections

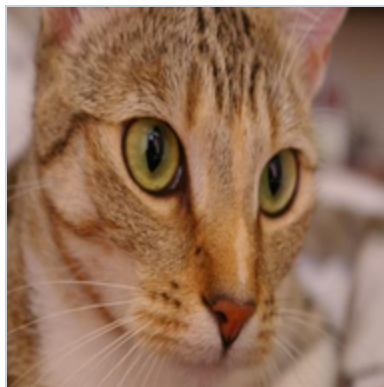
PASS

cat.jpg (no person) → NN returns 0 detections
count=0 NN=25.1ms

T09.3



input (192×192)



kit detections

PASS

T09.4 coffee.jpg (no person) → NN returns 0 detections
count=0 NN=25.1ms

PASS



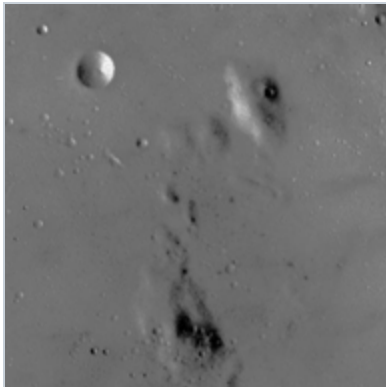
input (192×192)



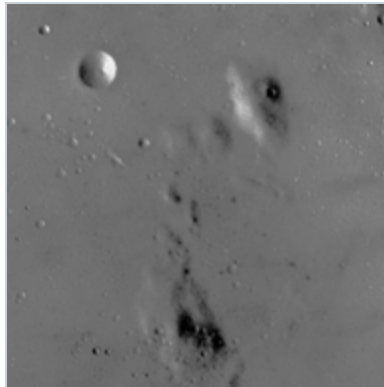
kit detections

moon.jpg (no person) → NN returns 0 detections
count=0 NN=25.1ms

T09.5



input (192×192)



kit detections

PASS

rocket.jpg (no person) → NN returns 0 detections
count=0 NN=25.1ms

T09.6



input (192×192)



kit detections

PASS

T09.7

NN inference avg ≤ 30 ms over 6 images
avg=25.1ms worst=25.1ms n=6

PASS

T10.1

`camera awb auto` accepted

PASS

T10.2

`camera exposure 10000` accepted

PASS

T10.3

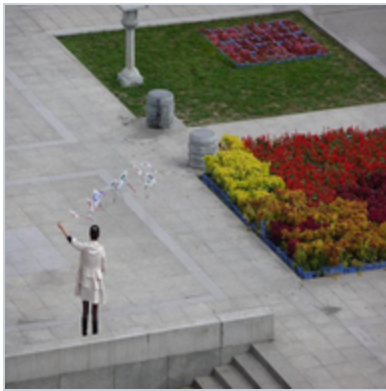
`camera gain 0` accepted

PASS

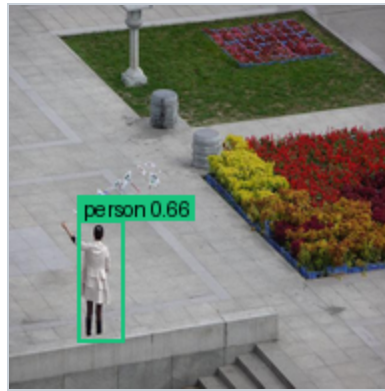
T11.1

1_person.jpg (single) → ≥ 1 person
detected: person=1 vehicle=0 (gt 1p/0v) NN=25.1ms

PASS



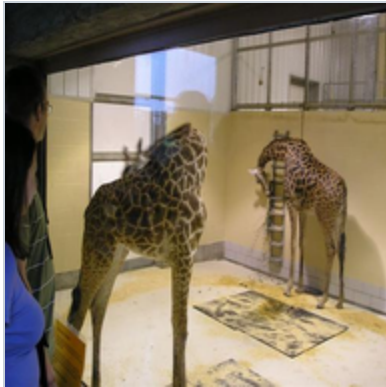
input (192×192)



kit detections

2_people.jpg (small group) → ≥1 person
 detected: person=2 vehicle=0 (gt 2p/0v) NN=25.1ms

T11.2



input (192×192)



kit detections

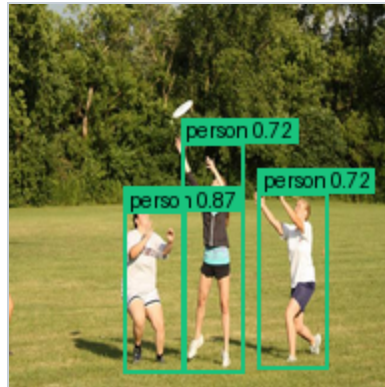
PASS

3_people.jpg (small group) → ≥2 person
 detected: person=3 vehicle=0 (gt 3p/0v) NN=25.1ms

T11.3



input (192×192)



kit detections

PASS

T11.4 5_people.jpg (group) → ≥2 person
 detected: person=2 vehicle=1 (gt 5p/0v) NN=25.1ms

PASS



input (192×192)



kit detections

7_people.jpg (group) → ≥2 person
detected: person=8 vehicle=0 (gt 7p/0v) NN=25.1ms

T11.5



input (192×192)



kit detections

PASS

T11.6 crowd_13.jpg (gt: 13p, 0v — crowd)

SKIP —
rc=-5

cars_18.jpg (vehicle (multi-class only)) → ≥0 person
detected: person=0 vehicle=1 (gt 0p/18v) NN=25.0ms

T11.7



input (192×192)



kit detections

PASS

T11.8 cars_15.jpg (vehicle (multi-class only)) → ≥0 person
detected: person=0 vehicle=2 (gt 0p/15v) NN=25.0ms

PASS



input (192×192)



kit detections

13ppl_5trucks.jpg (mixed) → ≥3 person
detected: person=7 vehicle=2 (gt 13p/5v) NN=25.0ms

T11.9



input (192×192)



kit detections

PASS